NUWANGA WIJAMUNI

AI Engineer

Negombo, Sri Lanka | +94 77 170 4614 | nuwanga69@gmail.com
linkedin.com/in/nuwanga-wijamuni-3bb4b8221 | github.com/Nuwanga-Wijamuni

PROFESSIONAL SUMMARY

AI Engineer developing agentic workflows, retrieval-augmented generation (RAG), and natural language-to-SQL applications. Experienced in Python, FastAPI, LangChain, LangGraph, vector databases, and AWS deployment, with an emphasis on agent tracing and human-in-the-loop approval.

TECHNICAL SKILLS

Agentic AI: LangGraph workflow orchestration, tool calling, structured LLM outputs, conversational memory, LangSmith tracing, API integration, human-in-the-loop approval

Generative AI: Large language models (LLMs), RAG, LangChain, prompt and context engineering, QLoRA fine-tuning, Hugging Face Transformers, PEFT, NVIDIA NIM

Programming and Data: Python, SQL, FastAPI, Flask, Streamlit, WebSockets, Pandas, SQLAlchemy, PostgreSQL, MySQL, Snowflake, Weaviate, ChromaDB

Cloud and MLOps: AWS (SageMaker, ECR, S3, RDS), Google Cloud (Vertex AI, Cloud Run), Docker, Kubernetes, KServe, CI/CD with GitHub Actions, MLflow, Prometheus, Grafana

ML and Automation: Scikit-learn, OpenCV, MediaPipe, Airflow, Power BI, Asana, Google Drive

PROFESSIONAL EXPERIENCE

HSB Marine Constructions Oct 2025 – Present

AI Engineer (Contract)

- Developed an internal RAG application with retrieval APIs, a Streamlit interface, and Docker Compose for local deployment; documented setup and troubleshooting.
- Built and tested an eight-stage executive task-triage workflow integrating ChatGPT, Google Drive, and Asana, with task ownership, prioritization, and CEO approval.
- Validated source-to-task mapping with source links and missing-information flags; refined duplicate checks and completion-evidence requirements, and configured the agent for operational review.
- Designed Company Brain knowledge structures and evaluated an internal retrieval architecture using Vertex AI Search, Cloud Run, Gemini, and speech-to-text.
- Supported ChatGPT Business administration, technical troubleshooting, and company systems documentation.

One Billion Tech Aug 2024 – Jul 2025

Data Engineer Intern *Colombo, Sri Lanka*

- Built and deployed a RAG application using LLaMA 3.3 and Weaviate, with GitHub, Docker, Amazon ECR, and Amazon SageMaker for containerized deployment.
- Optimized Snowflake queries with materialized views and datetime clustering keys, reducing execution times for those queries to under four seconds.
- Developed a natural language-to-SQL application connecting LLaMA 3.3 to Snowflake through FastAPI, Groq, and SQLAlchemy.
- Implemented stored-procedure logging to track new data entries and support auditing and data-load visibility.

MAS Holdings Mar 2023 – Mar 2024

Data Scientist Intern *Colombo, Sri Lanka*

- Built and evaluated ARIMA, Prophet, and LSTM models for Walmart sales forecasting at MAS ACME.
- Automated sales-data pipelines for training and validation; developed Power BI dashboards for MAS Kreeda and Bodyline.

SELECTED PROJECTS

Skyward Flight Search and Booking Agent
Python, FastAPI, LangGraph, Gemini API, React, TypeScript, SQLAlchemy
github.com/Nuwanga-Wijamuni/flight-booking-agent

- Built a flight-search application with a React interface and FastAPI backend; orchestrated request extraction and validation with LangGraph, optional Gemini structured outputs, and a Duffel sandbox adapter.
- Implemented quote expiry, user confirmation, and idempotent booking simulation, with SQLAlchemy and SQLite for persistence.

Agentic RAG System for E-Commerce
LangChain, LangGraph, LangSmith, Weaviate, SQL, Amazon S3

- Built an agentic retrieval system combining SQL data, vector retrieval, and object-storage sources to answer product queries.
- Orchestrated retrieval workflows with LangGraph and traced agent execution with LangSmith.

Gourmet Express Visual AI Agent
OpenAI Realtime API, React, WebSockets, FastAPI, Docker
github.com/Nuwanga-Wijamuni/AIVisualAgent

- Built a voice-controlled restaurant assistant using the OpenAI Realtime API and WebSockets; implemented tool calls to navigate the visual menu and update the shopping cart.

AI Customer Support Agent with Hybrid Memory
LangChain, ChromaDB, FastAPI, Docker

- Reduced token usage by over 70% in extended-dialogue project tests using hybrid conversational memory in a stateful customer-support agent.

GeoNIM Satellite Analyzer
Python, NVIDIA NIM, FastAPI, rasterio, GeoPandas, Google Cloud

- Built a pipeline for multi-gigabyte satellite imagery using image tiling and NVIDIA NIM vision-language models without local GPUs.
- Implemented structured JSON extraction to identify and map environmental anomalies, including deforestation.

NVIDIA DGX Spark for Company AI Workloads
Research project | AI infrastructure and multimodal data

- Evaluated DGX Spark for company PDFs, documents, images, audio, and video; researched MacBook connectivity, model-training and image-generation requirements, and hardware considerations.

CERTIFICATIONS AND TRAINING

AWS Certified Data Engineer – Associate (DEA-C01) 2024
NVIDIA Agentic AI training Completed

EDUCATION

Sri Lanka Institute of Information Technology (SLIIT)
Bachelor of Information Technology, Specialization in Data Science
Feb 2020 – Mar 2024 | Colombo, Sri Lanka
Relevant coursework: Data Engineering, Artificial Intelligence and Machine Learning, Cloud Computing

AWARDS

SLIIT Datathon 2022: Finalist, Merit Award for an AI-based monkeypox detection model
ML Nexus Quiz Competition: Second Runner-Up, University of Colombo

LANGUAGES

English, Sinhala, German